

Xiaohang Yu

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PROFILE

Agentic AI research at Imperial College London, building AI agents on trust and for trust: agent security, cryptographic protocols so an agent can safely be handed real authority; agentic forensics, agents put to work for society, blockchain forensics first; and RL post-training for stronger reasoning. PhD candidate; twice a startup co-founder (SafeClaw, 2026; a verifiable-AI infrastructure venture, 2023).

EDUCATION

Ph.D. Computer Science	Imperial College London, UK	2025 – 2027
Advised by William Knottenbelt. Research: agentic AI, agent security, agentic forensics, and RL post-training.		
M.S. Security Informatics	Johns Hopkins University, US	2016.08 – 2018.01
Focus: practical cryptography, applied machine learning, and network security.		
B.S. Computer Science	Beijing Normal University, China	2012.09 – 2016.06

RESEARCH

- **Agent Security**
Formalized the Agent Secret Use problem and designed a credential-delegation protocol that provably satisfies seven required security properties; the cryptographic foundation of SafeClaw.
Xiaohang Yu (first author), et al. *SUDP: Secret-Use Delegation Protocol for Agentic Systems*. arXiv preprint (under submission).
- **Agentic Forensics**
An agentic framework for blockchain forensics; PhD work.
Xiaohang Yu (first author), William Knottenbelt. *LOCARD: An Agentic Framework for Blockchain Forensics*. **IEEE ICBC 2026**.
- **RL Post-Training**
Empirical study of scaling behaviors in RL post-training for mathematical reasoning; a survey of agentic RL for LLMs.
Zelin Tan, Hejia Geng, **Xiaohang Yu**, et al. *Scaling Behaviors of LLM Reinforcement Learning Post-Training*. **ACL 2026** (main).
Guibin Zhang¹, Hejia Geng¹, **Xiaohang Yu**¹, et al. *The Landscape of Agentic Reinforcement Learning for LLMs: A Survey*. **TMLR 2025** (¹co-first).

WORK EXPERIENCE

Co-Founder	SafeClaw (safeclaw.pro)	2026.03 – present
- Built SafeClaw, an Agent Identity Solution: security infrastructure that puts agents' secret use under human control, giving agents scoped, per-use, user-approved delegation instead of reusable secrets that carry the user's full authority. - Led the design of SUDP, the cryptographic protocol underneath (see Research): human approval becomes an unforgeable, single-use authorization bound to the exact operation, so reusable authority never crosses the agent boundary. - Shipped the entire production stack solo in about three months: a Rust egress proxy with inline secret injection, an end-to-end encrypted vault with multi-device sync sealed under the user's passkey, a web console and cloud control plane; launched at safeclaw.pro in July 2026, first seed users within a week; open source at github.com/SafeClaw-OSS/safeclaw.		
Technical Co-Founder	Verifiable-AI Infrastructure Startup	2023 – 2025
Research and implementation of core protocols for verifiable AI systems and zero-knowledge-proof (ZKP) infrastructure; contributor to open-source standards including opML (Optimistic Machine Learning) and ERC-7007.		
Head of Blockchain Security	Chaitin Tech	2018.03 – 2023.03
Founded and led the blockchain security practice from zero, one of the earliest teams of its kind; tech lead and PM.		

- Protocol and cryptographic security audits across major L1s; discovered ~100 0-days across consensus, P2P, and verification layers; 5 CVEs in smart-contract audits.
- Technical lead for China's first national blockchain vulnerability scoring standard.
- Independent research: adapted JOP (Jump-Oriented Programming) to the EVM, first shown as a *Real World CTF 2018* challenge and later a canonical blockchain-CTF category; invited talk at *DEF CON 27 Village*.

SELECTED TALKS

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| • IEEE ICBC 2026 “LOCARD: An Agentic Framework for Blockchain Forensics”, Brisbane, Australia | 2026 |
| • DEF CON 27 Village “EVM Opcode JOP”, Las Vegas, US | 2019 |

QUALIFICATIONS & AWARDS

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| • 3rd Place, DEF CON CTF World Finals, team <i>Tea Deliverers</i> | 2021 |
| • Offensive Security Certified Professional (OSCP) | 2017 |
| • Project Management Professional (PMP) | 2021 |
| • Fully Funded Master Scholarship, China Scholarship Council (CSC) | 2016 |